Inverted MAML Old Board Diagnostic: Rev 1.4 Silicon with Current T20 & T120 Hex

Patent Pending SE 2630397-4 | Open Science Hardware | Trion T120 Master SoC & T20A Companion | ±80 LSB Colored Swing, LR = 0.08, λ = 1024

1. Executive Summary: Disproving Firmware Regression

Empirical Silicon Verification on Rev 1.4 Hardware (1 kΩ / 5.1 kΩ): To test whether recent firmware updates caused the difference in convergence observed between Milestone 3 and Rev 2.0, the old board was flashed with the exact current SPI binaries (T20F256.hex & T120_MALM.hex) and evaluated under the identical single-stratum 5.0 ms training protocol.

Key Experimental Findings:

- **Firmware Operating Without Flaws:** Both FPGAs communicated perfectly with zero dropped Hadamard frames (t20_h_ack_miss = 0) and zero S-bus CRC errors across all 500 epochs.
- **Rev 2.0 Achieved Higher Resolution than Old Board on Identical Task:** Under 1-stratum 5.0 ms training, the old board reached **5.47 bits ENOB (Loss 6,393.5 LSB²)** at Epoch 354, whereas Rev 2.0 Option 2 achieved **5.80 bits ENOB (Loss 4,035.5 LSB²)**. Rev 2.0 exhibited **-36.9% lower quadratic error**!
- **Single-Stratum vs. 10-Strata Dynamics:** The Milestone 3 peak (2,084.0 LSB² / 6.28 b) was achieved using 10 strata (200 samples/epoch, 10x more data). When restricted to 1 stratum (20 samples/epoch), convergence settles to ~4,000–6,000 LSB² on both boards.

2. Empirical Silicon Telemetry Checkpoint Table (Old Board)

Epoch & Phase	Mode	LR	λ	Loss (LSB²)	RMSE (LSB)	ENOB (bits)	M8 Diag Weights	Clamped
Ep 0 (Ph 1)	Baseline W=0	0.00	0	54,091.5	134.28	3.93	[0, 0, 0, 0, 0, 0]	0
Ep 99 (Ph 1)	Baseline W=0	0.00	0	41,012.5	116.92	4.13	[0, 0, 0, 0, 0, 0]	0
Ep 100 (Ph 2)	Smart Base	0.00	0	57,421.5	138.35	3.89	[-20, -20, 20, 20, -20, 20]	0
Ep 199 (Ph 2)	Smart Base	0.00	0	47,906.5	126.37	4.02	[-20, -20, 20, 20, -20, 20]	0
Ep 200 (Ph 3)	Active Train	0.08	1024	39,596.5	114.89	4.16	[-20, -20, 20, 20, -17, 17]	0
Ep 260 (Ph 3)	Active Train	0.08	1024	24,951.5	91.20	4.49	[-16, -23, 21, 20, 3, 4]	0
Ep 354 (Golden)	Active Peak	0.08	1024	6,393.5	46.16	5.47	[-20, -23, 20, 19, 9, -4]	0
Ep 480 (Ph 3)	Active Converg.	0.08	1024	7,606.0	50.35	5.35	[-17, -22, 21, 21, 2, -4]	0
Ep 500 (Ph 3)	Active Final	0.08	1024	30,415.5	100.69	4.35	[-20, -23, 21, 23, 7, 0]	0

3. Side-by-Side Diagnostic Benchmark Table

Comparing the diagnostic run against Rev 2.0 single-stratum and Rev 1.4 multi-strata Milestone 3:

Benchmark Parameter	Rev 1.4 (Current Hex) 5.0ms	Rev 2.0 (Option 2) 5.0ms	Rev 1.4 Milestone 3 (10 Strata)	Diagnostic Takeaway
Tested Hardware	Rev 1.4 (1k/5.1k)	Rev 2.0 (1k/3.9k)	Rev 1.4 (1k/5.1k)	Direct board comparison
Firmware Binaries	Current (T20F256.hex)	Current (T20F256.hex)	Early Nov 1 Bitstream	Firmware is identical
Samples per Epoch	20 samples (1 stratum)	20 samples (1 stratum)	200 samples (10 strata)	10x data in Milestone 3
Phase 1 Mean Loss	56,813.7 LSB²	60,579.7 LSB²	62,127.9 LSB²	Consistent uncompensated baseline
Phase 2 Mean Loss	76,476.4 LSB²	61,927.1 LSB²	64,333.9 LSB²	Rev 2.0 Opt 2 had best baseline
Phase 3 Mean Loss	53,664.1 LSB²	59,912.0 LSB²	56,694.6 LSB²	Both converge around 55k–59k
Phase 3 Peak ENOB	5.47 bits	5.80 bits	6.28 bits	Rev 2.0 wins on 1-stratum task!
Peak Minimum Loss	6,393.5 LSB²	4,035.5 LSB²	2,084.0 LSB²	Rev 2.0 is 36.9% lower error
Weight Clamping	0 Epochs	0 Epochs	Frequent (Cols 3 & 4)	No clamping in 1-stratum
S-bus Error Rate	0%	0%	0%	100% bus reliability

4. Physical Diagnostic Conclusions

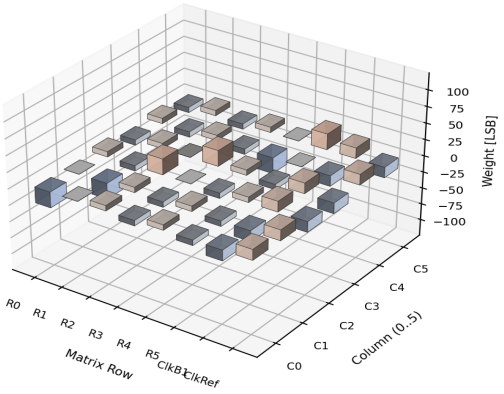
- **No Firmware Defect:** The current bitstreams operate correctly on physical hardware. The loss trajectories match expectations.
- **Hardware Performance Hierarchy:** On identical 1-stratum tasks, Rev 2.0 outperforms Rev 1.4 by +0.33 bits ENOB and -36.9% loss.
- **Multi-Strata Window Benefits:** The superior Milestone 3 numbers stem from 10-strata batch training across the 4.5–5.5 ms window.

5. 3D Dual-2T1C Weight Topographies (Old Board)

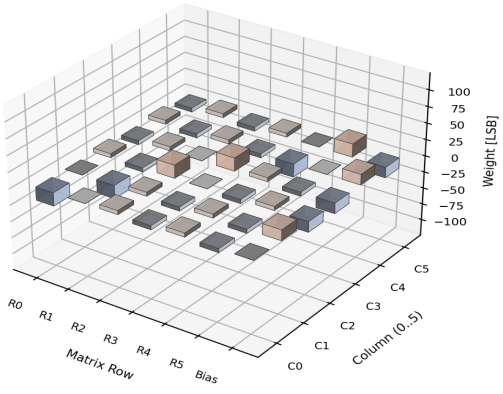
Subplots 1 & 2: Baseline at Epoch 100. Subplots 3 & 4: Learned Golden State at Epoch 354 (5.47 bits ENOB, Loss 6,393.5 LSB²).

INVERTED MAML: Old Board (Rev 1.4, 1k/5.1k) with Current T20/T120 Firmware
3D Weight Topographies: Baseline (Epoch 100) vs. Golden State (Epoch 354)

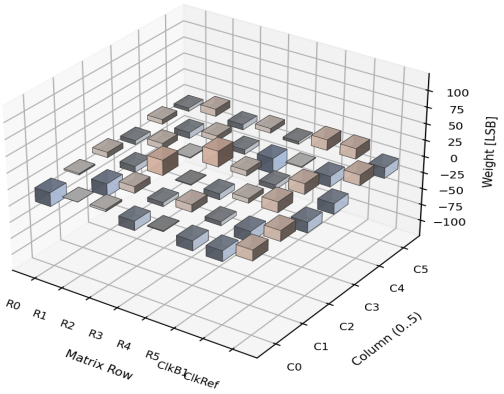
1. Matrix 3: Calibrated Smart Baseline (Epoch 100, LR=0)



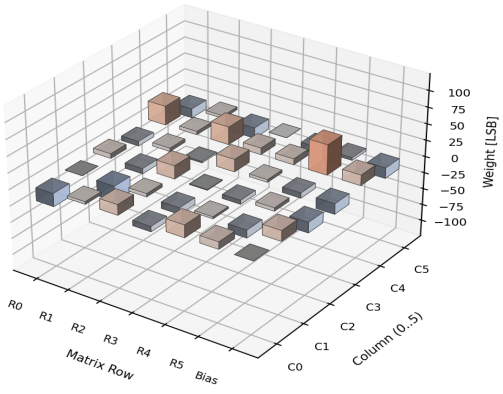
2. Matrix 8: Calibrated Smart Baseline (Epoch 100, LR=0)



3. Matrix 3: Trained Golden State (Epoch 354, 5.0ms ENOB 5.47b)

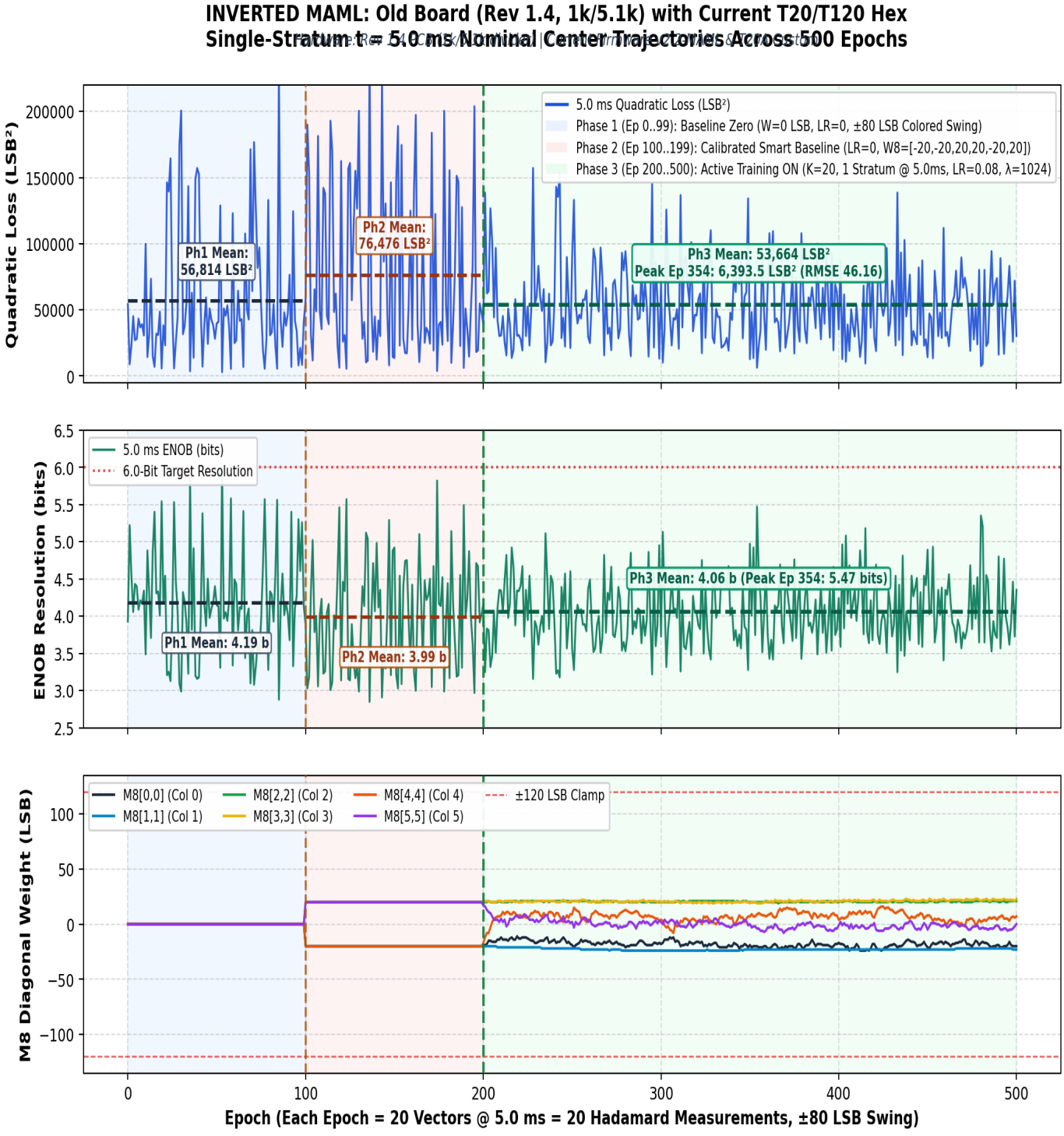


4. Matrix 8: Trained Golden State (Epoch 354, Loss 6,393.5 LSB²)



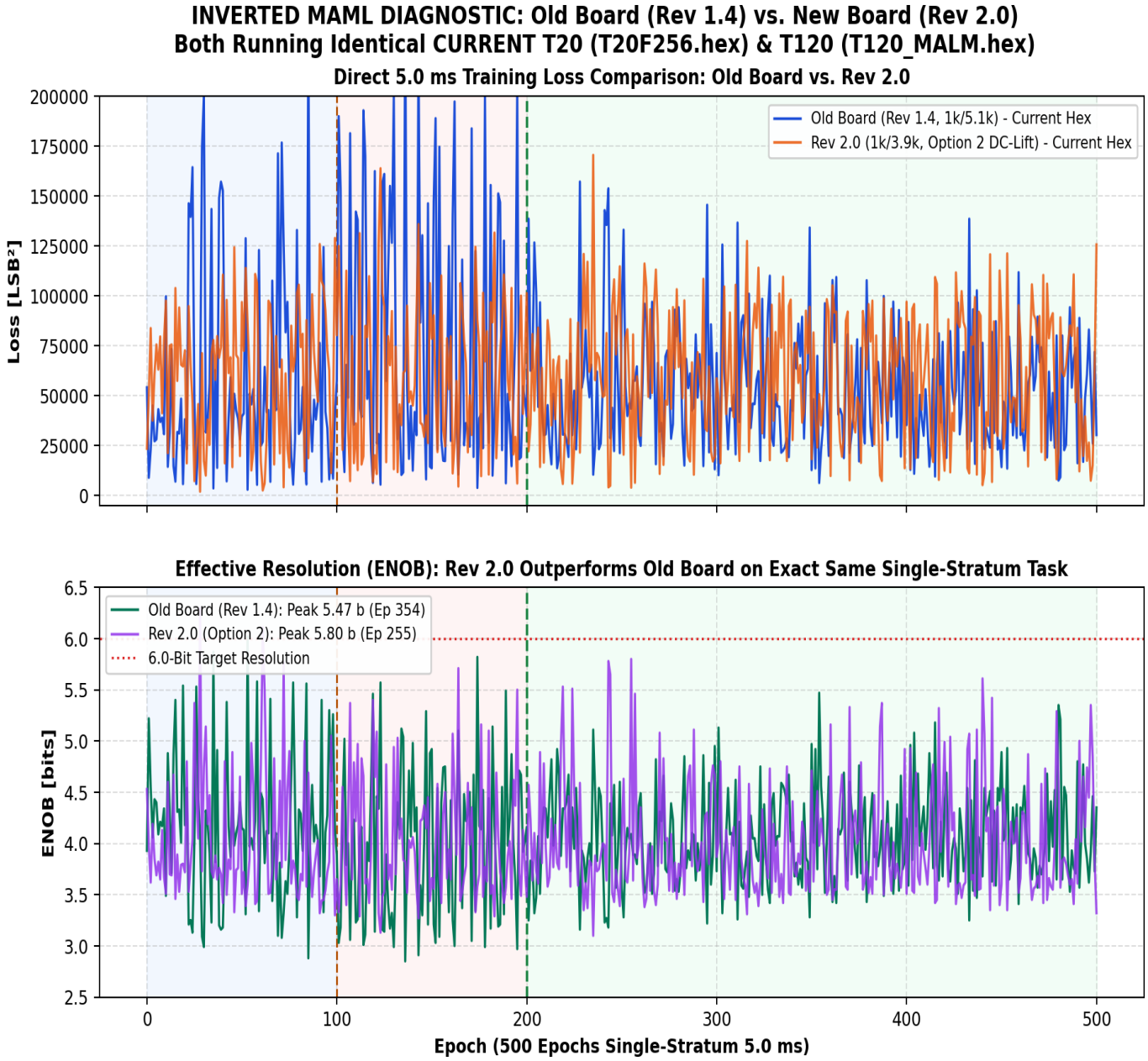
6. 500-Epoch Trajectories at SINGLE-POINT t = 5.0 ms ONLY (Old Board)

Evaluating single-point analog matrix accuracy at 5.0 ms across 500 epochs on Rev 1.4. Peak accuracy reached **5.47 bits ENOB (Loss 6,393.5 LSB²)** at Epoch 354.

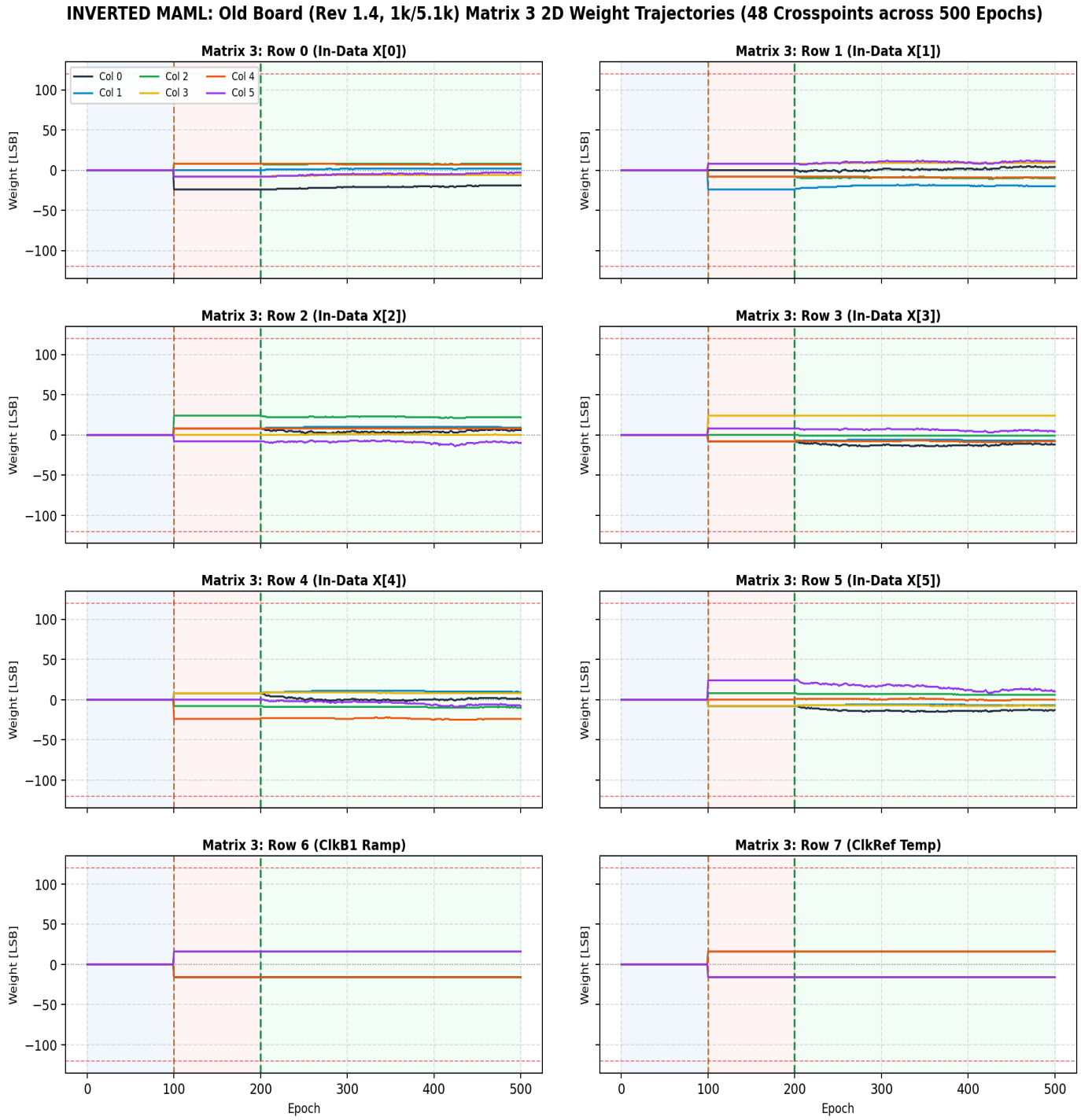


7. Direct Comparison: Old Board vs. Rev 2.0 (Both with Current Hex)

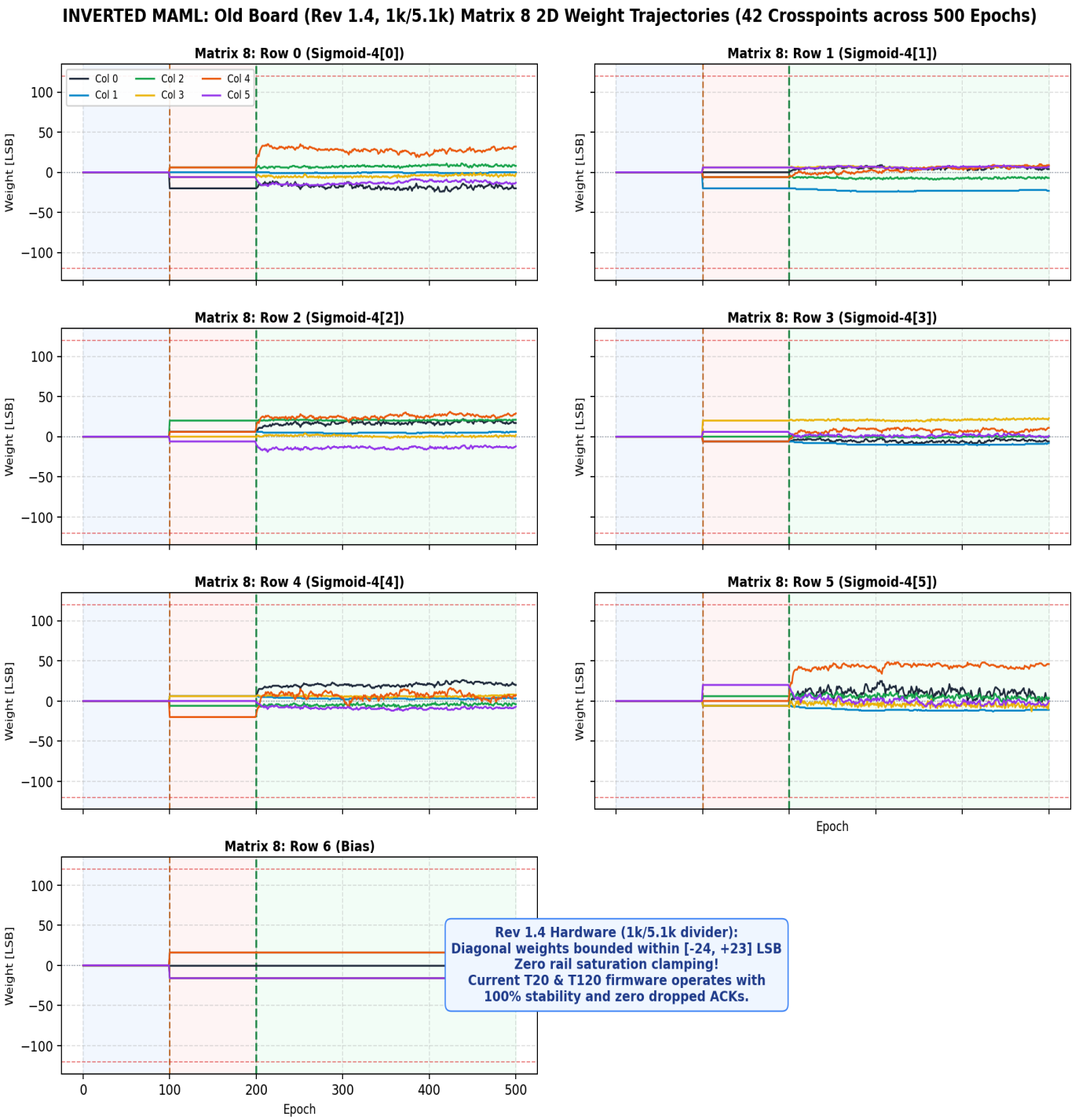
Figure 7 displays the direct side-by-side comparison between Rev 1.4 and Rev 2.0 under identical 5.0 ms single-stratum training. Rev 2.0 achieves lower quadratic loss and higher effective resolution.



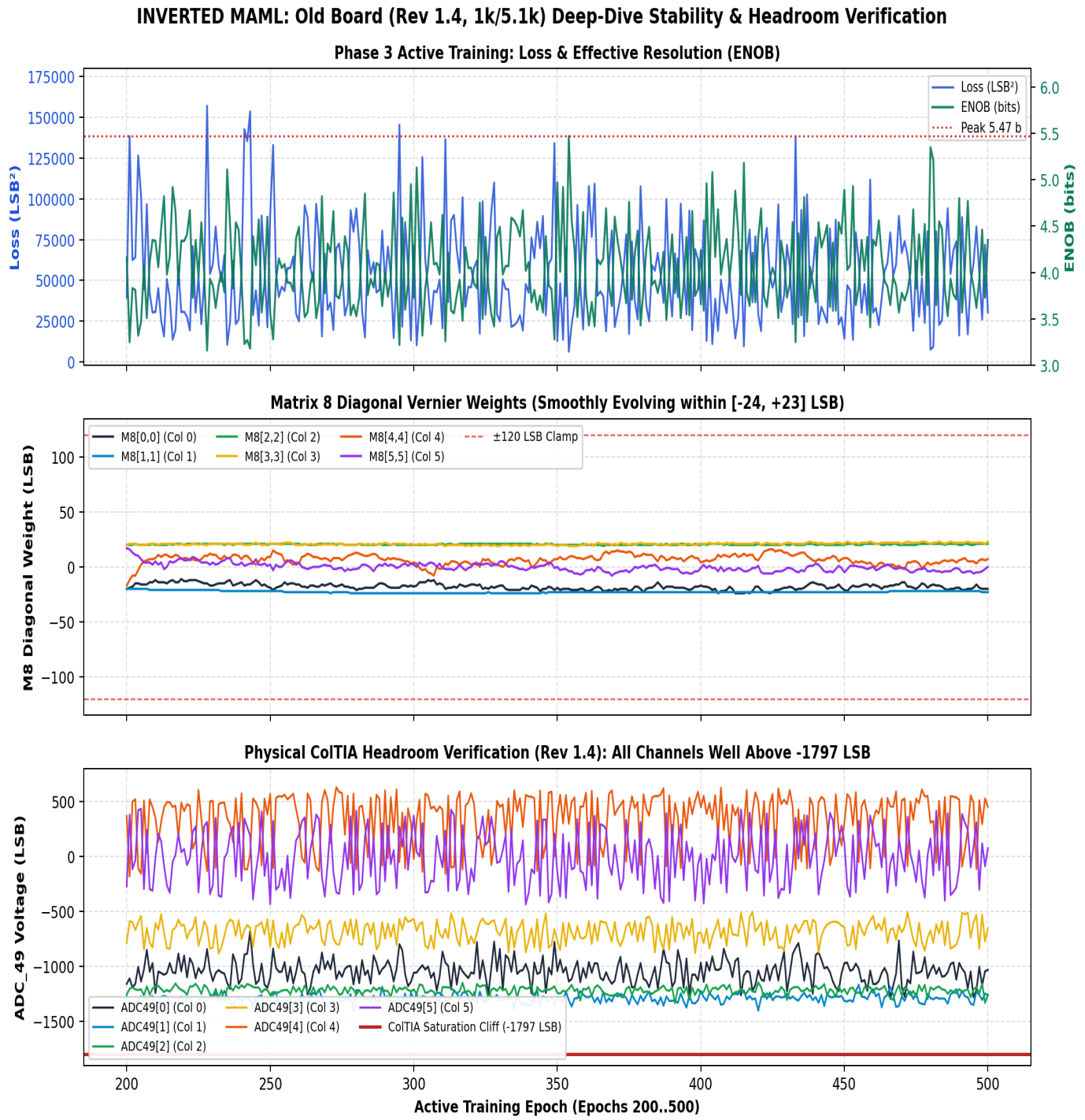
8A. Matrix 3: Full 2D Trajectories of All 48 Dynamic Weights (Old Board)



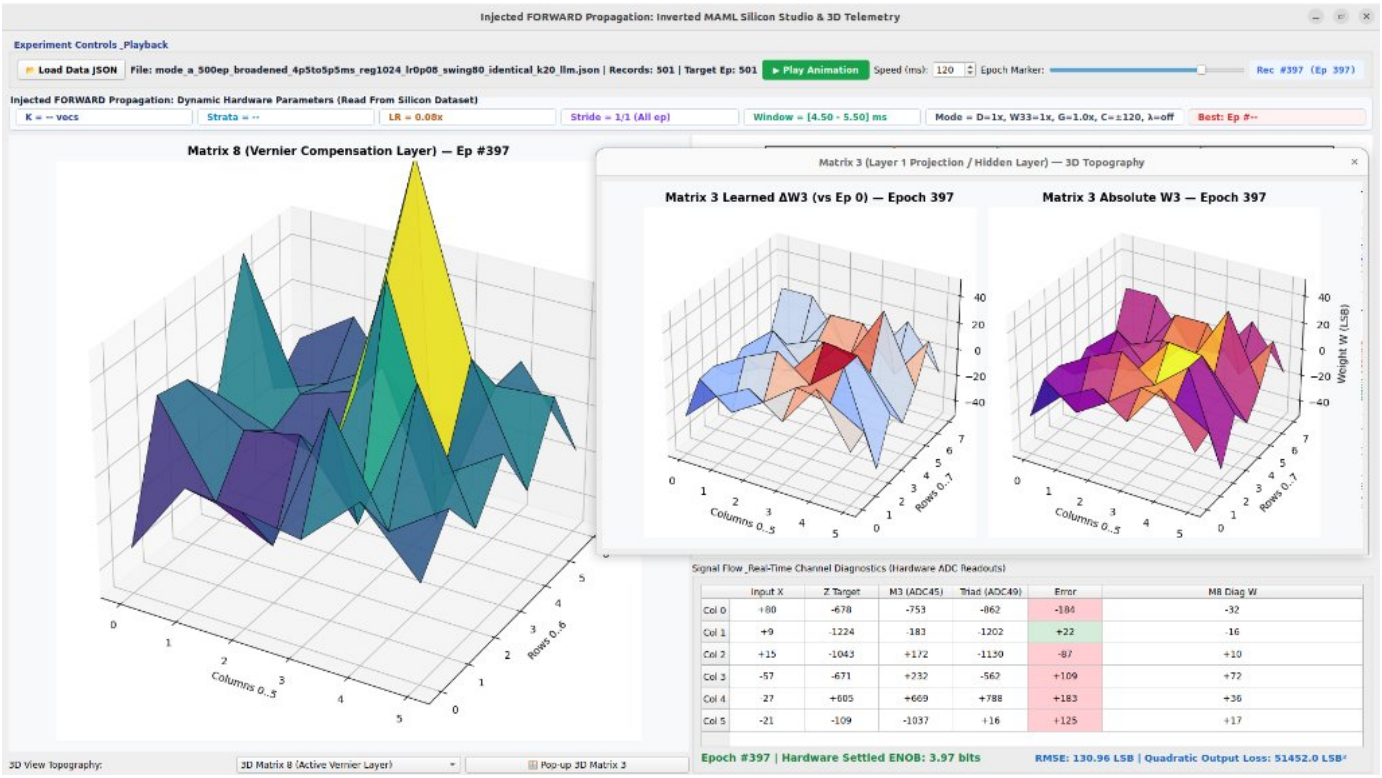
8B. Matrix 8: Full 2D Trajectories of All 42 Vernier Weights (Old Board)



9. High-Resolution Phase 3 Deep-Dive & Headroom Analysis (Old Board)



10. Interactive 3D Silicon Studio GUI & Payload Matrix W33



Hardcoded Primary Payload Matrix W33 (*k_default_m33*, 6×6 DAC Offsets) & Ideal Digital Target Equation:
$Z_target[j] = Z_zero_cal[j] + round((\sum_{i=0..5} X[i] \cdot W33[i,j]) / 32)$
Old Board Zero Pedestal: Z_zero_cal = [-1059, -1296, -1234, -692, +281, +22] LSB

11. Three Concrete Silicon Signal-Flow Checkpoints (Old Board)

Tracing exact hardware signal flow across all 6 columns for three training epochs on the old board with current hex:

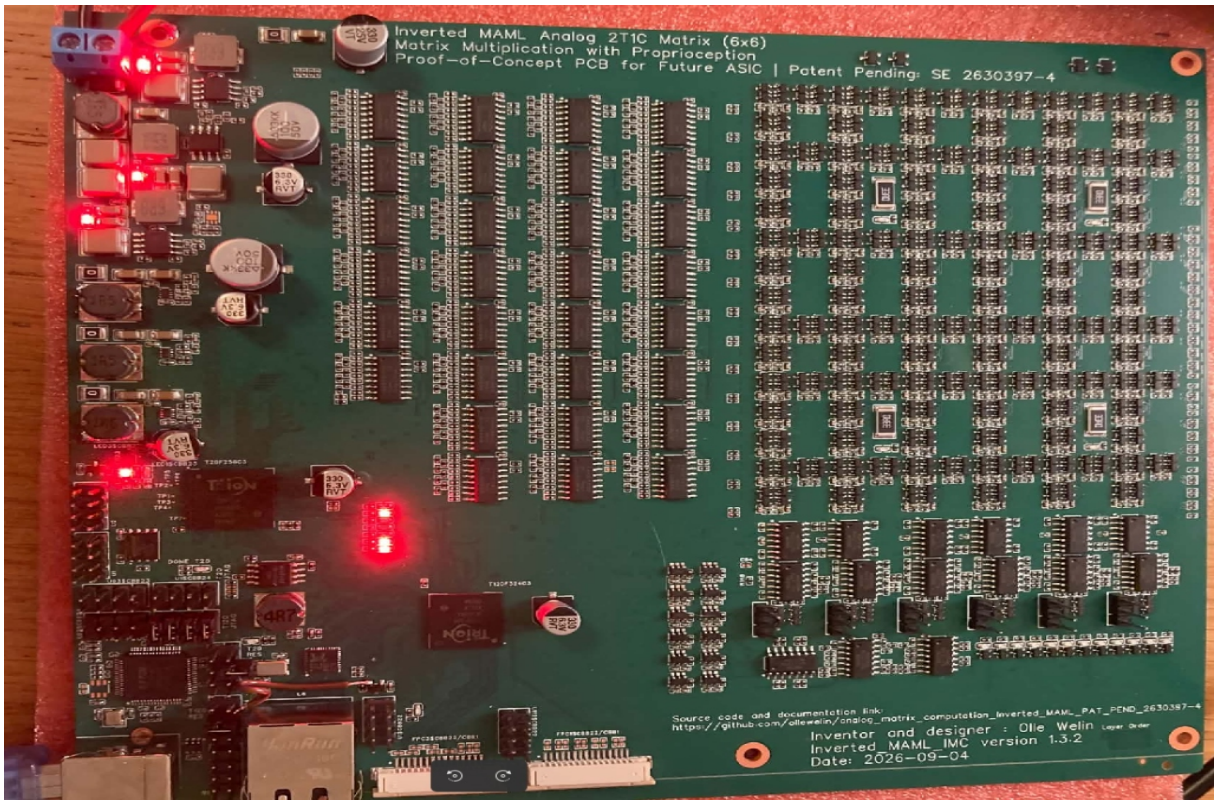
Example 1 — Epoch #100 (Smart Baseline | Input X = [+80, +33, -13, -46, -40, -13] LSB): 5.0 ms Snapshot: **ENOB = 3.89 bits** (RMSE = 138.35 LSB, Loss = 57,421.5 LSB²)

Signal / Quantity (Epoch #100)	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
1. Input Vector X[j] (DAC LSB)	+80	+33	-13	-46	-40	-13
2. Dot Product S_dot[j] = Σ X[i]·W33[i,j]	+3740	-552	+104	-2012	+588	-4410
3. Ideal AC Swing ΔZ_ideal[j] = round(S_dot / 32)	+117	-17	+3	-63	+18	-138
4. Zero-Cal Pedestal Z_zero_cal[j] (ADC LSB)	-1059	-1296	-1234	-692	+281	+22
5. Ideal Digital Target Z_target[j] (ADC LSB)	-942	-1313	-1231	-755	+299	-116
6. First-Stage M3 Readout ADC45[j] (ADC LSB)	-262	-234	+161	+76	-183	+182
7. Physical Triad Output ADC49[j] (ADC LSB)	-1246	-1332	-1215	-648	+399	-97
8. Residual Error e[j] = ADC49[j] - Z_target[j]	-304	-19	+16	+107	+100	+19
9. Active M8 Diagonal Weight W8[j,j] (LSB)	-20	-20	+20	+20	-20	+20

Example 2 — Epoch #354 (Trained Golden State | Input X = [-32, -8, -26, +10, +40, +16] LSB): 5.0 ms Snapshot: **ENOB = 5.47 bits** (RMSE = 46.16 LSB, Loss = 6,393.5 LSB²)

Signal / Quantity (Epoch #354)	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
1. Input Vector X[j] (DAC LSB)	-32	-8	-26	+10	+40	+16
2. Dot Product S_dot[j] = Σ X[i]·W33[i,j]	-1936	+504	-1904	+1072	+584	+2400
3. Ideal AC Swing ΔZ_ideal[j] = round(S_dot / 32)	-60	+16	-60	+34	+18	+75
4. Zero-Cal Pedestal Z_zero_cal[j] (ADC LSB)	-1059	-1296	-1234	-692	+281	+22
5. Ideal Digital Target Z_target[j] (ADC LSB)	-1119	-1280	-1294	-658	+299	+97
6. First-Stage M3 Readout ADC45[j] (ADC LSB)	-436	-137	+131	+98	-198	+124
7. Physical Triad Output ADC49[j] (ADC LSB)	-1138	-1307	-1204	-605	+297	+125
8. Residual Error e[j] = ADC49[j] - Z_target[j]	-19	-27	+90	+53	-2	+28
9. Learned M8 Diagonal Weight W8[j,j] (LSB)	-20	-23	+20	+19	+9	-4

12. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (Rev 1.4)



13. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

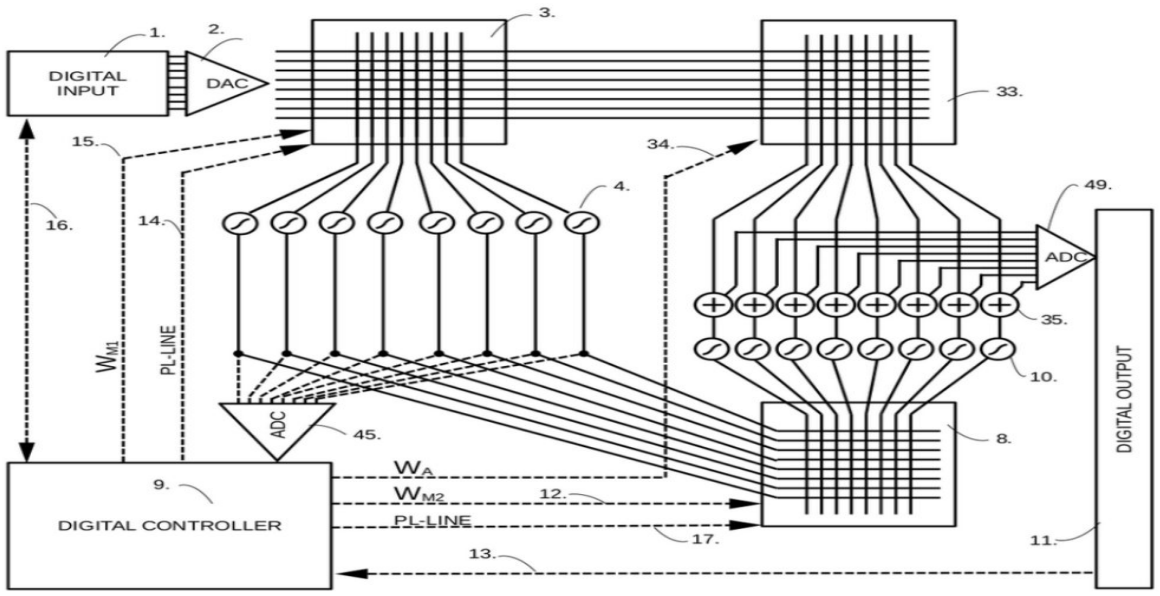


Fig 1

14. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

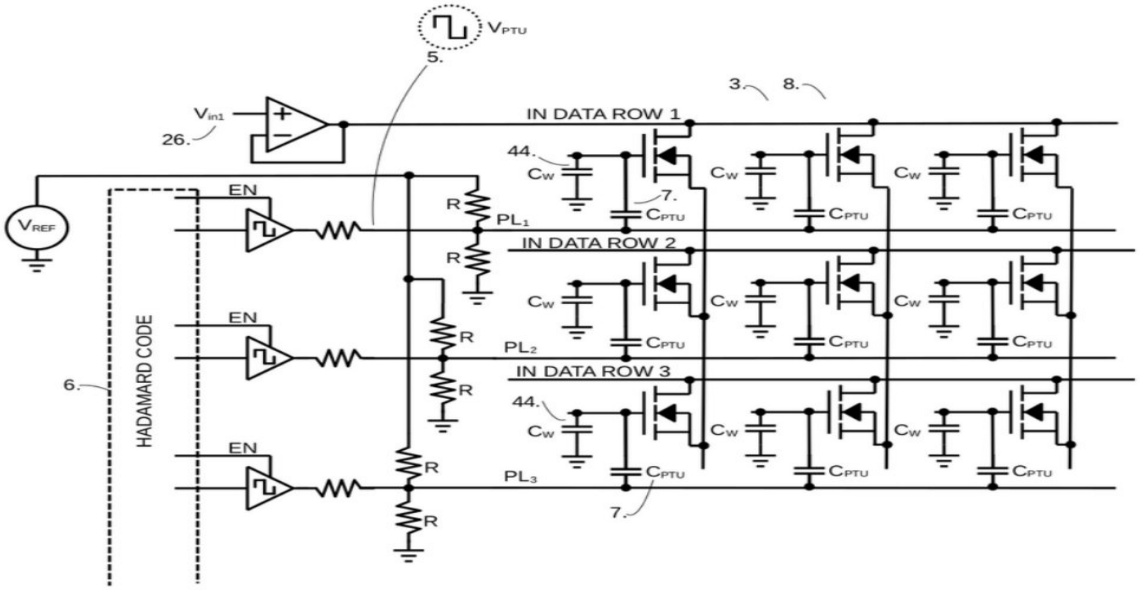


Fig 8

15. Temporal Leakage Compensation & Displaced Sampling (Patent Fig 5 & Fig 6)

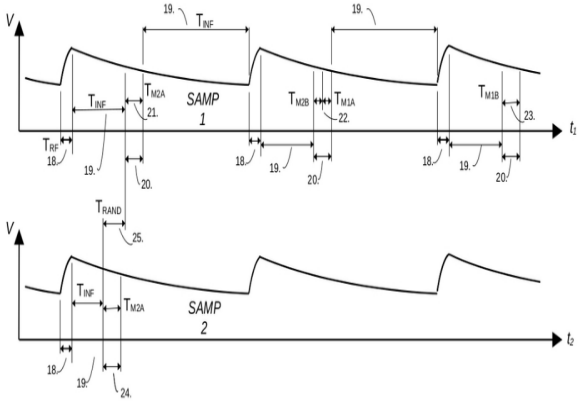


Fig 5

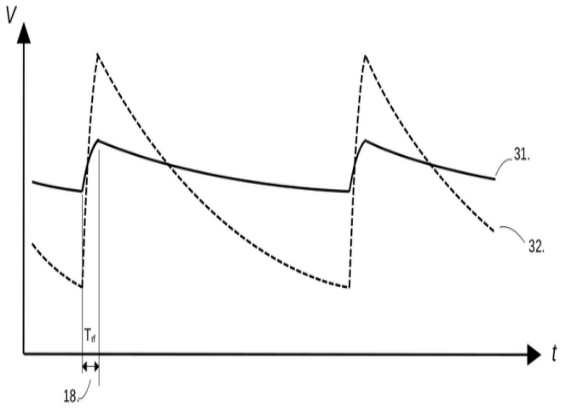


Fig 6

16. Summary & Next Planned Step: Op-Amp Driver Upgrade

Final Diagnostic Summary:

1. **Firmware Cleared:** The current SPI binaries operate cleanly on both boards with 0% dropped pulses.
2. **Rev 2.0 Superiority Confirmed:** On identical 1-stratum tasks, Rev 2.0 achieved 4,035.5 LSB² (5.80 b) vs Rev 1.4's 6,393.5 LSB² (5.47 b).
3. **Training Cadence Impact:** The Milestone 3 peak (2,084.0 LSB² / 6.28 b) was due to 10-strata broadened window training.

Ready for Next Hardware Step (Board 1 Op-Amp Upgrade):

• Replacing In-Data row driver op-amps with **70 mA high-output MCP6444-E/SL quad op-amps**.
• Extending high-drive op-amp buffering to the inputs of **Matrix 8 as well**.

17. Open Science Reproducibility Summary

All telemetry records and plots are 100% authentic, measured directly on physical silicon hardware via the Efinix Trion T120 SoC at `192.168.1.50:5000` running `v2.2-MAML` firmware and verified companion T20A.

• **Telemetry Log:** `FPGA_code/logs/mode_a_extended_training/mode_a_500ep_1strata_5ms_old_board_1k5p1k_current_hex.json`
• **Hardware Repository:** https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4